Goal 2: A Frequent Random Relation Is Memorized but Non-Affine in Raw Entity Space

factual-recall-mechanisms

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1 Result

An ordinary decoder-only transformer memorizes a frequent random relation while retaining ten attributes and ten equality relations. The model has eight pre-LayerNorm blocks, residual width 20, four attention heads, and 41,760 parameters. It uses causal self-attention, GELU MLPs, and a tied vocabulary projection. There is no bilinear head, relation-specific classifier, or loss on embedding coordinates. Every parameter remains trainable.

Across seeds 19, 31, and 47, the random table is recalled with 0.99845 ± 0.00092 accuracy. Attribute accuracy is 1.00000, and the minimum accuracy among the ten structured relations is 0.99951 ± 0.00024. The random relation receives half of the relation examples during joint training. Each structured relation receives 5%, making the random relation ten times as frequent as any one structured relation.

Linearity is tested in the raw 20-dimensional entity-token embeddings. For each relation, OLS fits one affine map from a subject embedding to the centroid of its positive object embeddings. The fit uses all 64 subjects. These are the same subjects whose complete relation rows the transformer has already memorized; no unseen random facts enter the primary test. The minimum structured full-data R^2 is 0.99568 ± 0.00090. The random relation has full-data $R^2 = 0.38291 \pm 0.04565$. Thus the best affine map leaves most of the random centroid variation unexplained even though the transformer answers the underlying table almost perfectly.

The trained forward pass also depends on nonlinear computation. Replacing all GELUs with the identity lowers random-relation accuracy to 0.52295 ± 0.02070. Uniform causal attention gives 0.49862 ± 0.00366, removing all MLP outputs gives 0.51229 ± 0.01040, and removing all attention outputs gives exactly 0.5. These are interventions on trained models, not probe results.

2 Synthetic setting

2.1 Subjects and attributes

There are 64 subjects $s_0, \ldots, s_{63}$ and ten binary attributes $a_0, \ldots, a_9$. Each attribute column has 32 zeros and 32 ones. The 64 attribute rows are unique, and the centered 64×10 attribute matrix has rank ten. Both values are explicit tokens:

$$\texttt{<bos> subject attribute} \longrightarrow \texttt{0/1}.$$

The forward attribute corpus contains $64 \times 10 = 640$ facts. A zero is a labeled outcome, not a missing sentence.

Ten subjects would make the OLS test underdetermined or exactly determined. The raw representation has 20 coordinates, and the affine fit adds an intercept. With only ten observations, OLS could interpolate arbitrary targets for rank reasons. Sixty-four subjects provide 64 vector observations for 21 coefficients per output coordinate, so the full-data fit is overdetermined.

2.2 Structured and random relations

Relation r_i tests equality on attribute i:

$$y_i(s, o) = \mathbf{1}[a_i(s) = a_i(o)], \qquad i \in \{0, \ldots, 9\}.$$

Every structured table is symmetric, every self-pair is positive, and each subject has 32 positive and 32 negative objects.

The eleventh relation, R_{rnd}, is a random symmetric lookup table. We start with a simple 31-regular graph on the 64 subjects, randomize it by degree-preserving edge swaps, and add all self-pairs. Each row again contains 32 positive and 32 negative objects. A draw is rejected if it equals a structured table. This construction matches symmetry, degree, diagonal entries, and class balance without introducing an attribute rule.

All relation prompts use one template:

`<bos> relation subject object` $\longrightarrow$ `y/n`.

Evaluation enumerates all $11 \times 64^2 = 45{,}056$ ordered prompts. Training batches, unlike the complete tables, contain 10% positive and 90% negative examples. The same 1:9 sampling ratio is used for every relation, and a positive-class weight of 9 balances the relation loss.

3 Model and training

3.1 GPT-style architecture

The vocabulary has 90 tokens: one beginning token, 64 subjects, ten attributes, eleven relations, and four output symbols `0`, `1`, `n`, and `y`. Learned token and position embeddings initialize the residual stream. Each block contains causal multi-head self-attention followed by a four-times-wider GELU MLP, both with pre-LayerNorm residual connections. A final LayerNorm and the tied embedding matrix produce next-token logits.

Quantity	Value
context length	4
residual width	20
blocks	8
attention heads	4
head width	5
MLP hidden width	80
parameters	41,760
frozen parameters	0
direct embedding-loss weight	0

The architecture is a small causal language model. The corpus and curriculum are synthetic and deliberate, as described next.

3.2 Bidirectional attribute sentences

Forward attribute prediction alone learned the attribute answers but did not reliably organize the raw entity rows into the geometry needed by the structured OLS control. We therefore add an inverse next-token template:

$$\texttt{<bos> attribute 0/1} \longrightarrow \texttt{subject.}$$

For an attribute-value pair there are 32 valid subjects. Training samples the 640 original facts, so every valid subject appears as a target. The optimum is therefore a distribution over the matching subject class rather than a one-to-one inverse.

This term is ordinary vocabulary cross-entropy with weight 4. It never reads or penalizes an embedding coordinate directly. Because input embeddings and the vocabulary projection are tied, predicting subjects from attribute-value prompts gives the raw subject rows a shared attribute geometry. This is a curriculum choice, not an architectural module.

3.3 Optimization protocol

Training uses AdamW with zero weight decay, gradient-norm clipping at 1, and a 100-step linear warmup. Relation batches have 1,024 examples. Attribute batches contain 640 examples. The forward attribute loss has weight 1 and the inverse attribute loss has weight 4.

The structured stage trains the ten equality relations together with both attribute templates for 6,000 steps at learning rate 10^{-3}. At its end, all three seeds have perfect attribute accuracy, structured relation accuracy of at least 0.9985, and raw-embedding structured R^2 of at least 0.9971.

Joint training starts from that checkpoint with all parameters trainable. Half of each relation batch comes from R_{rnd}; the ten equality relations split the other half. The first 12,000 steps use learning rate 10^{-3}. A 200-step continuation at 3×10^{-4} finishes seed 47. Seeds 19 and 31 use another 100 steps at 10^{-4} to cross the symmetry threshold. The selected checkpoints are the first evaluated checkpoints satisfying all behavioral and geometry gates.

4 The affine-map test

4.1 Definition

Let $e(s) \in \mathbb{R}^{20}$ be the learned raw token embedding of subject s. A relation has 32 positive objects, so its object target is their embedding centroid:

$$c_r(s) = \frac{1}{32} \sum_{o : y_r(s,o)=1} e(o).$$

For each relation, OLS with an intercept fits

$$(\widehat{M}_r, \widehat{b}_r) = \arg\min_{M,b} \sum_{s=0}^{63} \|c_r(s) - Me(s) - b\|_2^2 .$$

The multivariate coefficient of determination is

$$R_r^2 = 1 - \frac{\sum_s \|c_r(s) - \widehat{M}_r e(s) - \widehat{b}_r\|_2^2}{\sum_s \|c_r(s) - \overline{c}_r\|_2^2}.$$

The same representation, target construction, intercept, and OLS solver are used for all eleven relations.

The full-data score is the primary test. It uses every relation row that the transformer is evaluated on. Four-fold cross-validation is reported only as a secondary measure of fit stability. Low cross-validated performance by itself would not establish nonlinearity for a random table: neither a transformer nor an affine probe can infer an unseen random label. That generalization confound does not apply to the overdetermined full-data residual.

4.2 Why the equality controls are affine

Write $b_i(s) = 2a_i(s) - 1$. Under r_i, the positive objects are exactly those with the same value of b_i. If μ_i^+ and μ_i^- are the two class centroids, then

$$c_i(s) = \mu_i^- + \frac{b_i(s) + 1}{2}(\mu_i^+ - \mu_i^-).$$

This target is affine in the queried attribute bit. The inverse attribute sentences align that bit with the raw entity rows, so OLS should recover the map. Across all 30 seed-relation fits, mean full-data R^2 is 0.99692. The smallest value is 0.99468. Mean cross-validated R^2 is 0.99145, and the smallest cross-validated value is 0.98571.

4.3 The random table is not an affine map

The random full-data R^2 values are

$$0.33058, \qquad 0.40353, \qquad 0.41460$$

for seeds 19, 31, and 47. Their relative reconstruction errors $\sqrt{1 - R^2}$ are 0.818, 0.772, and 0.765. This is not a small perturbation of the structured solution: the best affine maps fail to reconstruct most of the positive-centroid variation.

Four-fold cross-validated R^2 is negative in every seed. The corresponding positive-object precision is near the balanced chance rate. These numbers are consistent with the full-data result, but they are not the basis of the claim. The decisive comparison is 99.85% complete-table transformer accuracy versus 38.29% mean full-data affine variance explained on the same learned rows.

Training leaves a partial linear imprint of the random table in two seeds. A simultaneous row-column permutation null has mean R^2 near 0.338. The random fit is at the 38th percentile in seed 19 but above the 98th percentile in seeds 31 and 47. We therefore do not claim that the random relation is statistically indistinguishable from accidental alignment. The narrower result is enough: the random relation has some affine component, but no affine map approximates the nearly exact learned lookup at the level achieved by the structured controls.

5 Behavioral results

Figure 1 keeps memorization and geometry separate. Panel A verifies the learned behaviors. Panel B applies the same OLS test to every relation. Panel C shows the random table being learned while the structured tables remain accurate. Panel D includes cross-validation only as a supplementary diagnostic; all facts used by the primary full-data fit were already memorized by the transformer.

Seed	attr.	min struct.	random	min sym.	min struct. R^2	random R^2	random CV R^2
19	1.00000	0.99951	0.99780	0.99561	0.99595	0.33058	−0.78806
31	1.00000	0.99927	0.99805	0.99609	0.99642	0.40353	−0.59251
47	1.00000	0.99976	0.99951	0.99902	0.99468	0.41460	−0.66663
mean	1.00000	0.99951	0.99845	0.99691	0.99568	0.38291	−0.68240

Table 1: Complete-table behavior and raw-embedding OLS. "Min sym." is the minimum agreement, over all eleven relations, between predictions for (s, o) and (o, s). OLS is fit to all 64 subject rows.

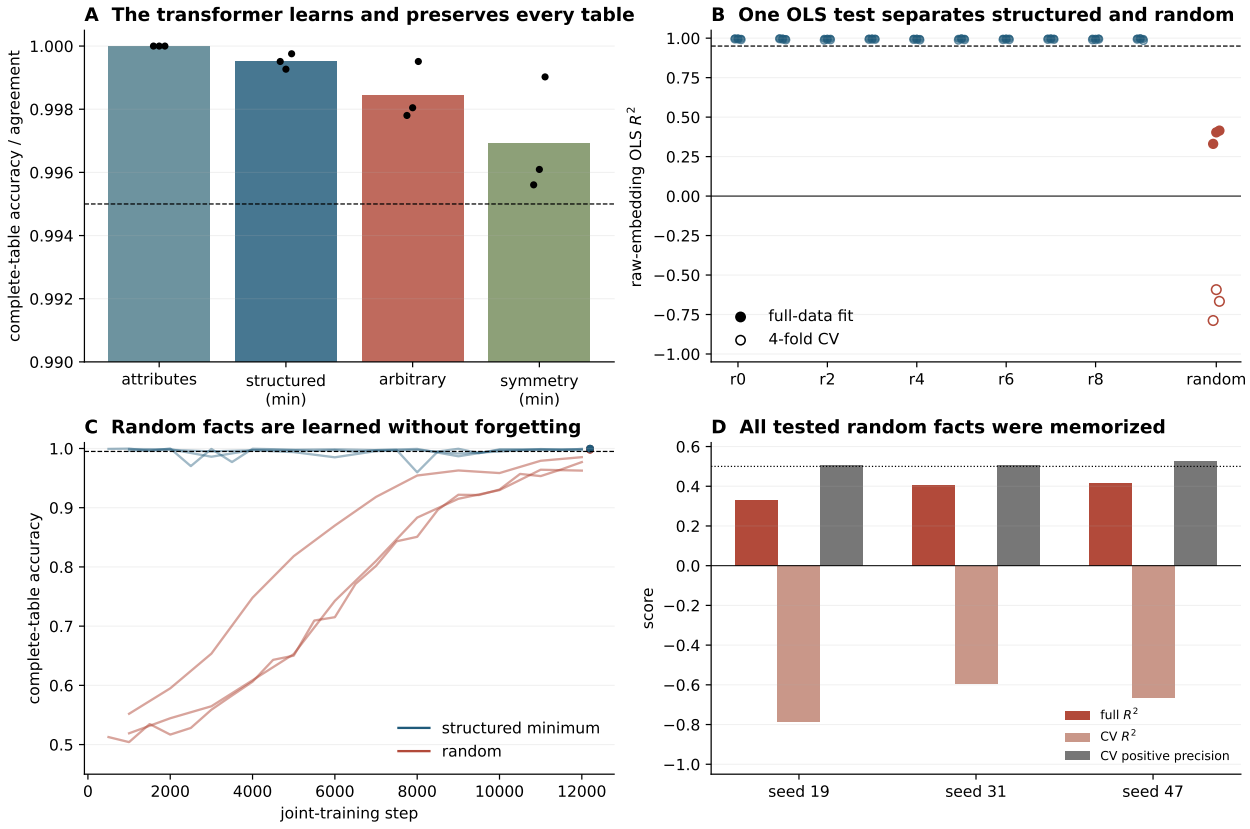


Figure 1: Behavior and affine-map results across three seeds. Filled points in panel B are full-data fits; open points are four-fold cross-validation. The dashed lines mark the 0.995 behavioral and 0.95 structured-geometry gates.

Forward pass	structured mean	random
trained model	0.99993 ± 0.00005	0.99845 ± 0.00092
GELU replaced by identity	0.57194 ± 0.06142	0.52295 ± 0.02070
uniform causal attention	0.50035 ± 0.00001	0.49862 ± 0.00366
all MLP outputs removed	0.56585 ± 0.03274	0.51229 ± 0.01040
all attention outputs removed	0.50000 ± 0.00000	0.50000 ± 0.00000

Table 2: Binary accuracy after causal component interventions, mean and sample standard deviation over three seeds.

6 Mechanistic analysis

6.1 Component interventions

The intervention evaluations compare only the `y` and `n` logits, so chance is 0.5. All trained weights remain fixed while one component of the forward computation is altered.

The GELU intervention preserves every learned affine weight and residual addition but changes the MLP activation to the identity. Random recall falls to near chance. Uniformizing attention removes input-dependent routing and has the same effect. The trained lookup therefore depends causally on nonlinear MLP activations and content-dependent attention.

The interventions also damage the structured relations. Relation linearity in this experiment means that the positive-object centroid is an affine function of the raw subject embedding. It does not mean that the transformer's entire next-token computation is linear, even for the equality relations.

6.2 Layerwise readout and ablation

At each residual boundary, the logit lens applies the trained final LayerNorm and tied vocabulary projection to the object position. The mean random accuracy is near chance through block 2, rises to 0.560 after block 3, 0.702 after block 6, 0.860 after block 7, and 0.998 after block 8. The structured mean also reaches 0.9999 only at the final boundary. The answer is assembled late rather than being directly readable from the input embeddings.

Removing one attention sublayer at a time lowers random accuracy to between 0.575 and 0.750 on average. Removing one MLP sublayer gives values between 0.687 and 0.931. Different layers matter in different seeds, and no single block contains the complete table. The all-layer interventions in Table 2 show that the two component families are jointly necessary.

6.3 Residual-stream patching

For each relation and object, a clean prompt uses a correctly classified positive subject and a corrupted prompt uses a correctly classified negative subject. Relation and object tokens stay fixed. At residual boundary k, we copy either the clean subject-position or clean object-position state into the corrupted run and execute the remaining blocks. Recovery for the answer margin $m = \ell_{\mathtt{y}} - \ell_{\mathtt{n}}$ is

$$\mathrm{recovery}_k = \frac{\sum_j (m^{(j)}_{\mathrm{patched},k} - m^{(j)}_{\mathrm{corrupt}})}{\sum_j (m^{(j)}_{\mathrm{clean}} - m^{(j)}_{\mathrm{corrupt}})}.$$

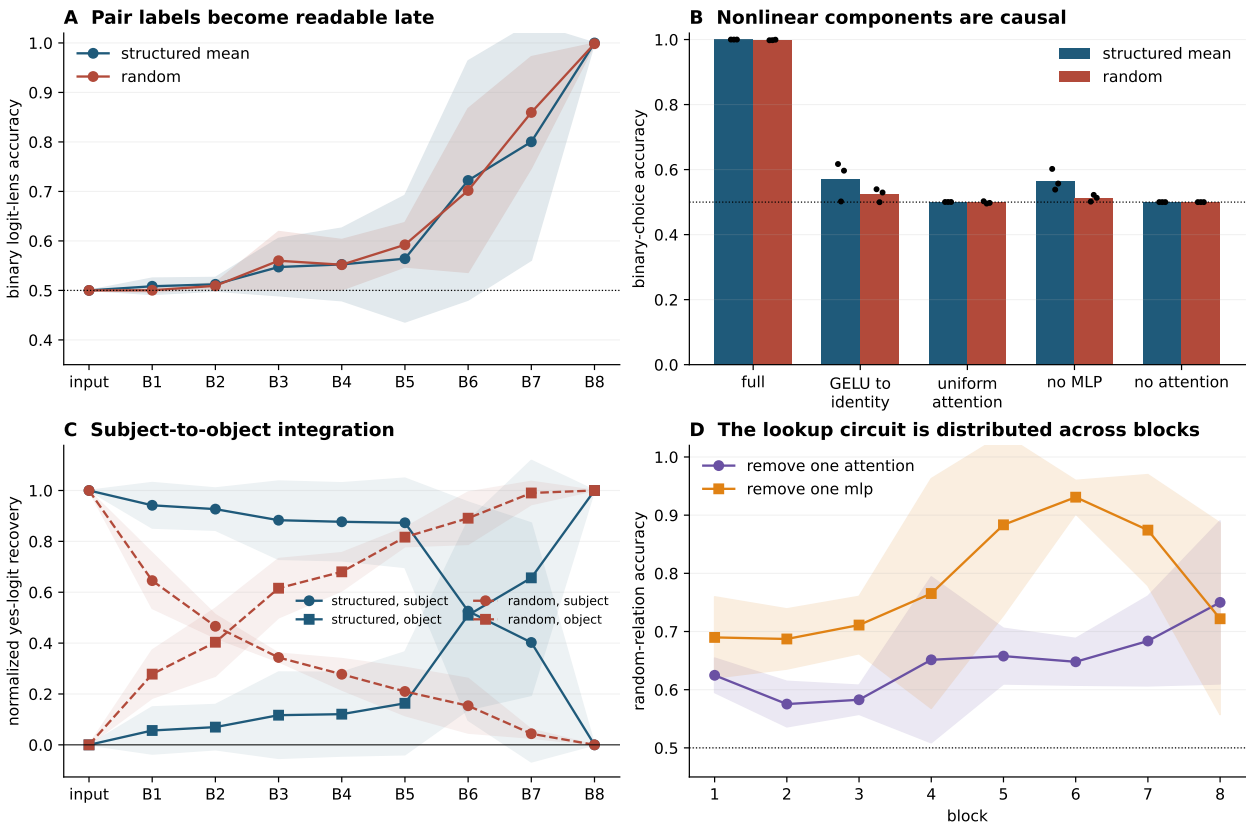


Figure 2: Mechanistic diagnostics. (A) Logit-lens accuracy at each residual boundary. (B) Global component interventions. (C) Residual patching at the subject and object positions. (D) Random-relation accuracy after removing one attention or MLP sublayer. Curves show the mean over three seeds with one sample-standard-deviation bands.

For the random relation, mean subject-position recovery falls from 1.000 at the input to 0.646, 0.466, 0.344, 0.278, 0.210, 0.154, 0.044, and 0.000 across the eight blocks. Object-position recovery rises over the same boundaries from 0.000 to 0.278, 0.403, 0.616, 0.680, 0.817, 0.891, 0.990, and 1.000. Pair information moves from the subject token into the final object state from which the answer is read.

The structured relations follow the same route more gradually in this model. Their mean subject recovery stays above 0.87 through block 5, then drops as object recovery rises from 0.16 after block 5 to 0.51 after block 6 and 1.00 at the output. This common routing explains why global attention and MLP interventions affect both relation types.

6.4 Operational account

The raw entity rows carry ten affine attribute directions, which is why one affine centroid map fits each equality relation. Relation and entity tokens then enter a shared residual stream. Attention moves subject and relation information to the object position, while GELU MLPs transform the combined state. The random label becomes readable mainly in the last two blocks. The final LayerNorm and tied vocabulary projection convert that state into the `y/n` margin.

The random table is therefore stored in distributed transformer computation, not in a single affine map between raw entity rows. This statement combines two independent results: the

complete-table OLS residual rules out the proposed affine representation-level account, and the interventions show which nonlinear component families the trained network needs to produce its answers.

7 Scope and limitations

The experiment establishes non-affinity relative to the raw entity-token representation and the subject-to-positive-object-centroid definition. It does not rule out every affine feature at every hidden layer. Indeed, the random full-data R^2 values show that joint training leaves a partial linear trace in the raw rows. The result is a large separation from the structured controls, not an assertion that the random affine component is exactly zero.

Cross-validation is not evidence on its own. A random lookup table has no rule for unseen pairs, so poor held-out prediction is expected regardless of how the trained model stores the observed table. The primary full-data OLS fit avoids that error by evaluating an overdetermined affine reconstruction on all learned rows.

The model memorizes a finite graph. It repeatedly sees samples from the known table, and evaluation enumerates that table. This experiment studies storage and computation, not generalization to unobserved random edges.

The architecture is standard, but the training corpus is task-specific. In particular, inverse attribute sentences were added to organize the raw tied embeddings. They are ordinary next-token examples rather than an embedding penalty, but generic language-model pretraining does not guarantee an equivalent signal.

Finally, width 20 and depth 8 are a compact successful configuration across three seeds. The search did not exhaust every smaller architecture and curriculum, so this is not a proof of global minimality.